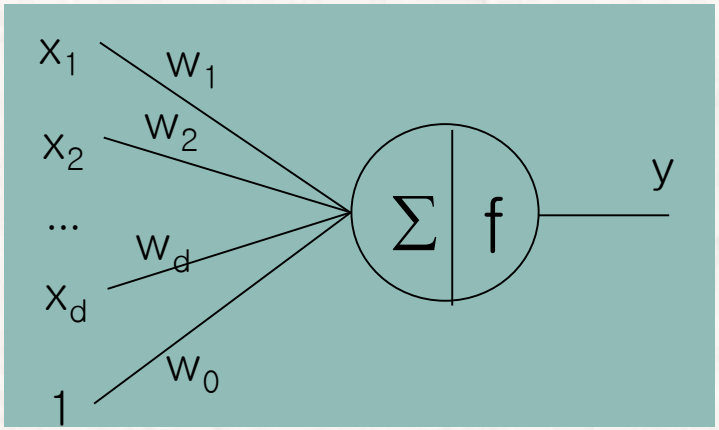
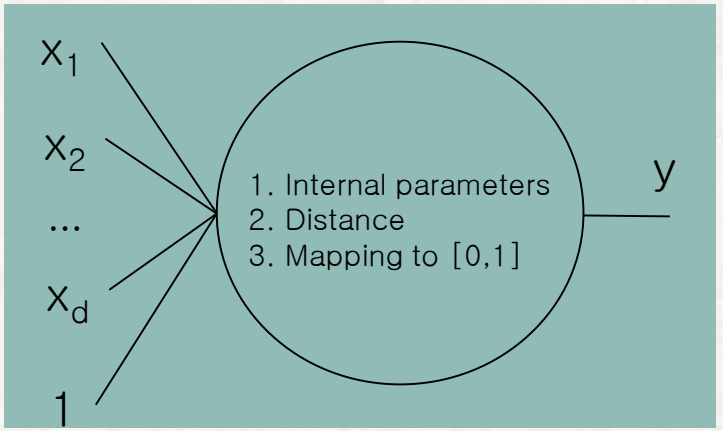
Radial Basis Function NN

Perceptron

- Perceptron

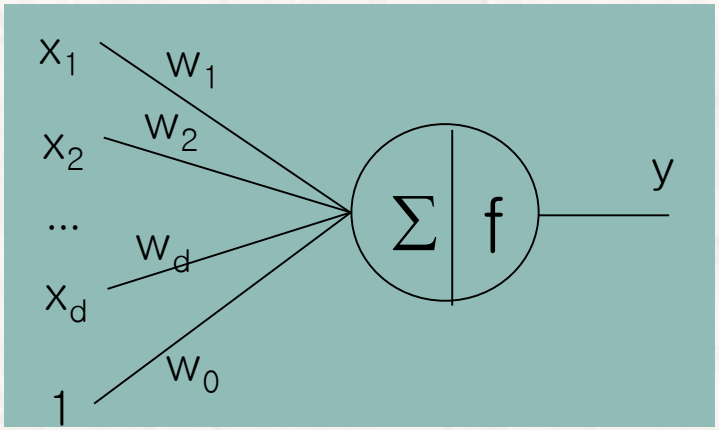


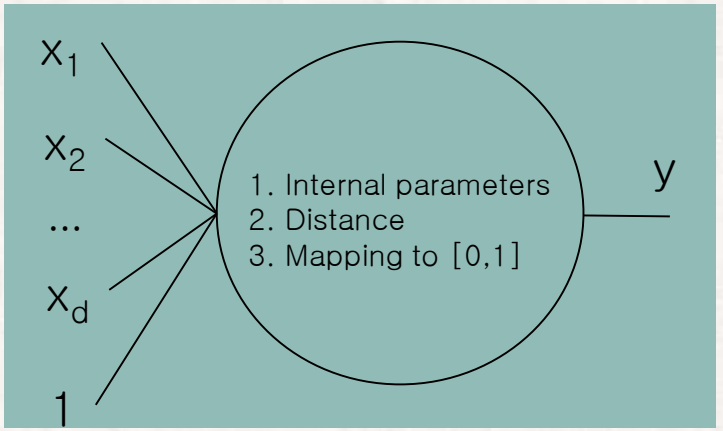


  - Internal Parameters: a given LINE
  - Distance: the distance between the LINE and the input
  - Mapping: if input is above the LINE, output 1
    if input is below the LINE, output 0

Perceptron

- Function of Perceptron



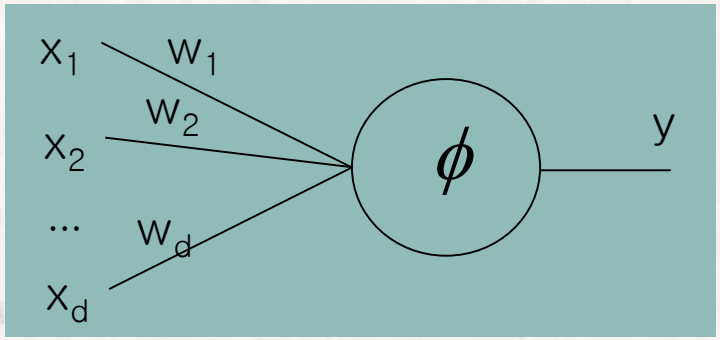


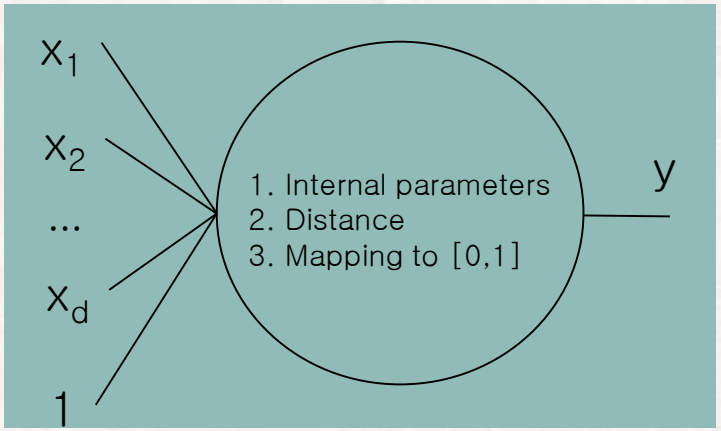
- The three steps in a equation: $y = \dfrac{1}{1+\exp\left(-\sum_{i=0}^{d} w_i x_i\right)}$

Let's consider another way of nonlinear mapping

Perceptron

- Node of another type

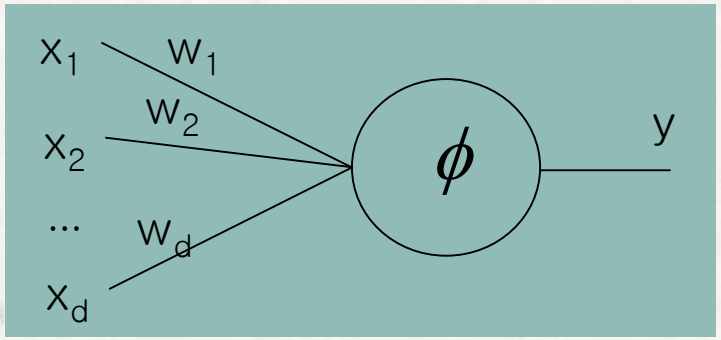


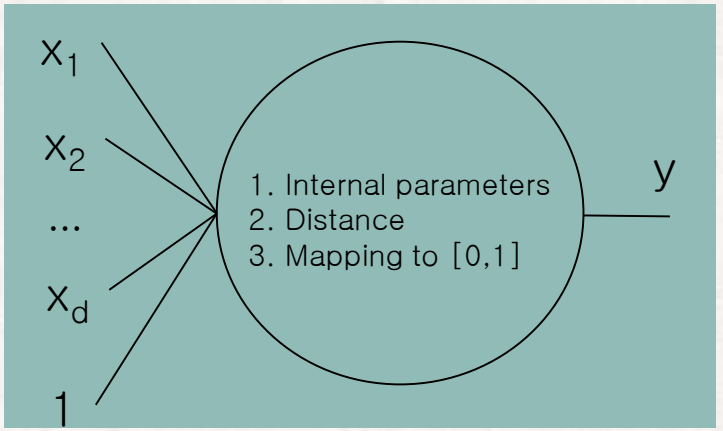


- Internal Parameters: a given POINT
- Distance: the distance between the POINT and the input
- Mapping: if input is close to the POINT, output 1
  if input is far from the POINT, output 0

Perceptron

- Node of another type



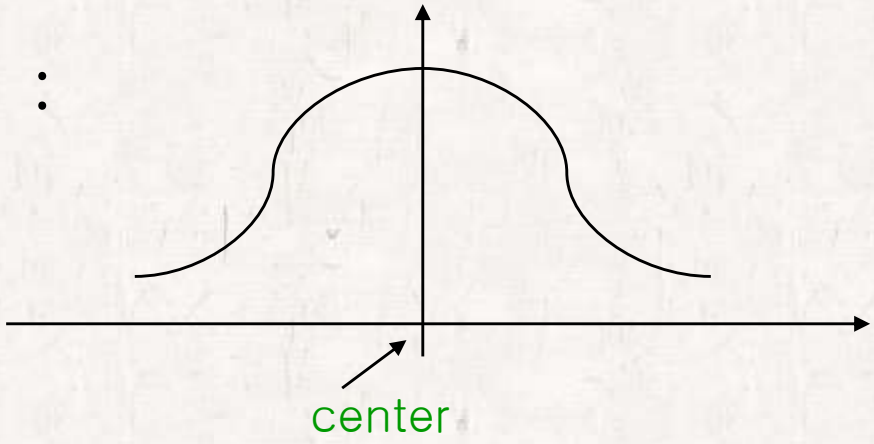


- The three steps in a equation: $$y = \exp\left(-\frac{(\mathbf{w}\cdot\mathbf{x}-\mathbf{m})^2}{2\sigma^2}\right)$$

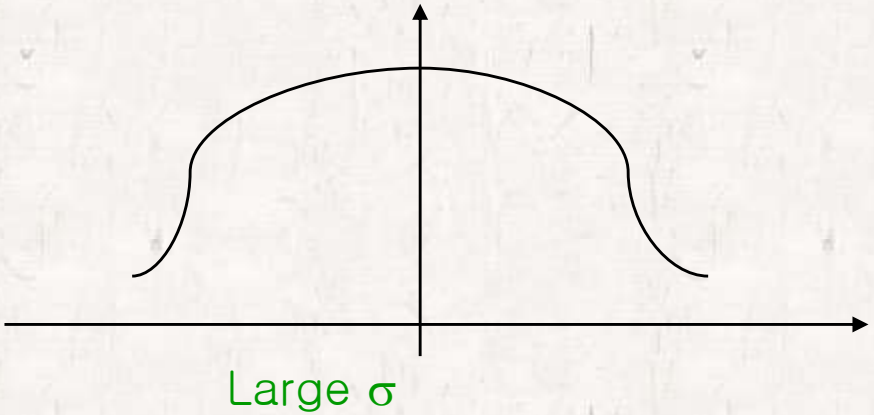
Radial-Basis Function

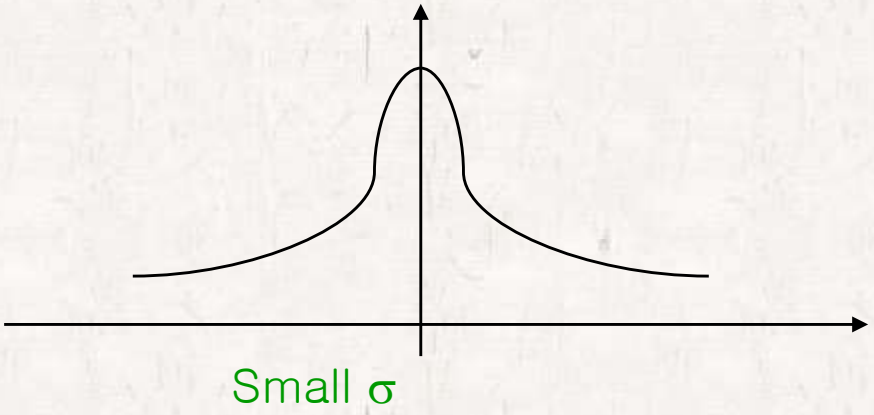
New Neurons

ϕ :



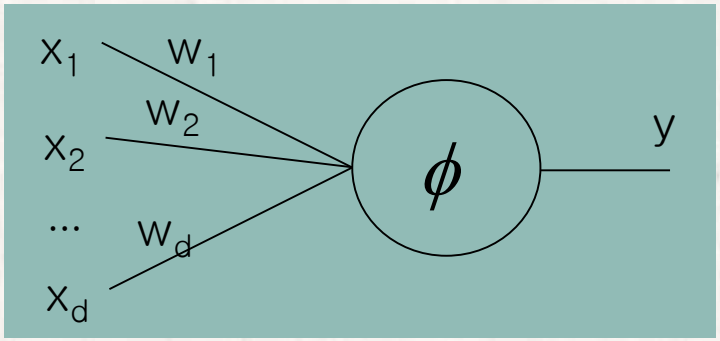
σ is a measure of how spread the curve is:





New Neurons

- Function of New Neurons



$$y = \varphi(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{w}\cdot\mathbf{x} - \mathbf{m}\|^2}{2\sigma^2} \right) = \exp\left(-\frac{1}{2\sigma^2} \sum_{j=1}^{d} \left(w_j x_j - m_j \right)^2 \right)$$

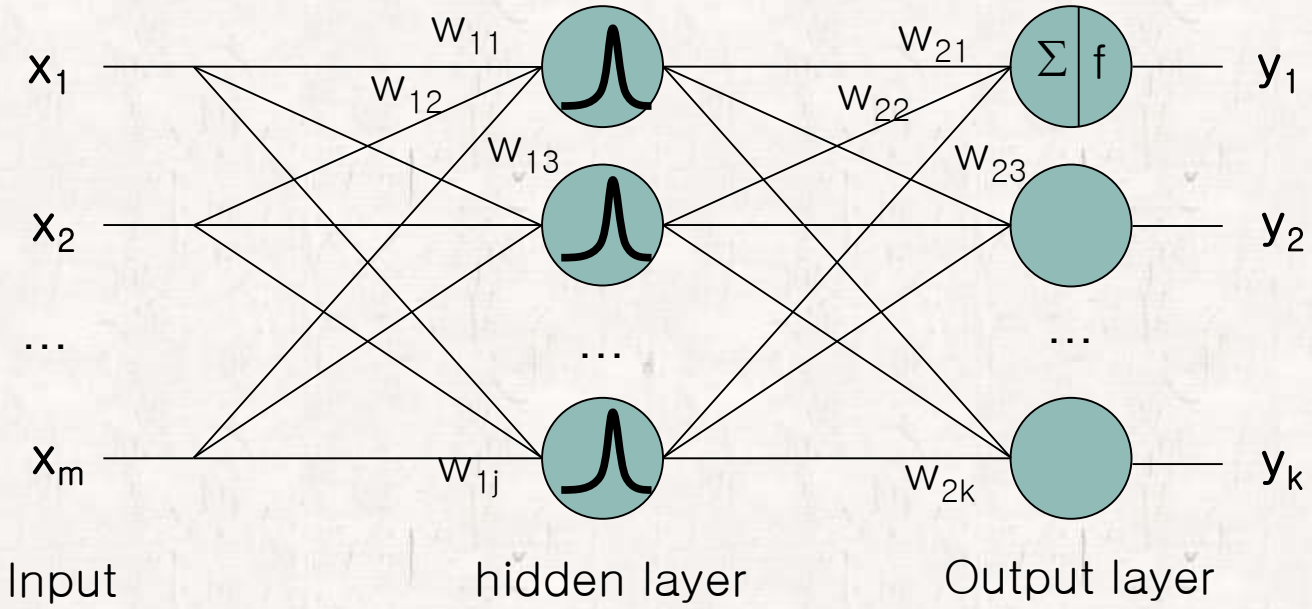
where $\mathbf{m}$ and σ are given

$$\mathbf{m} = \left(m_1, m_2, \cdots, m_d \right)$$

$$\sigma = \text{a real number}$$

Radial-Basis Neural Network

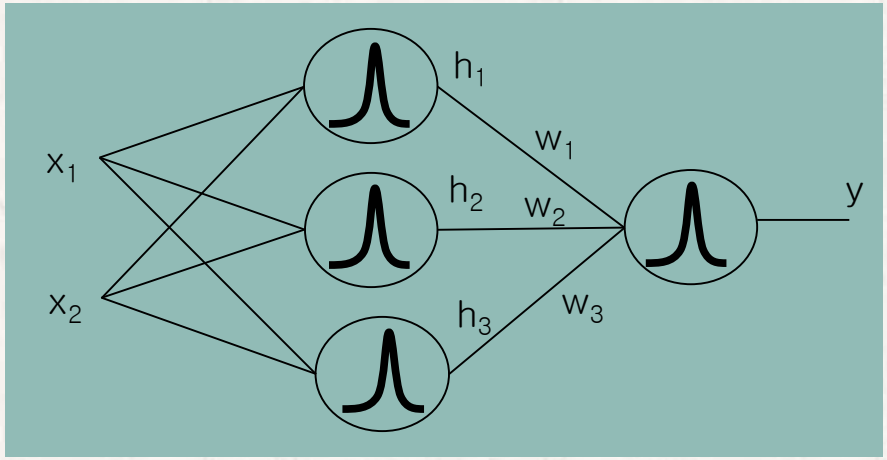
- Structure

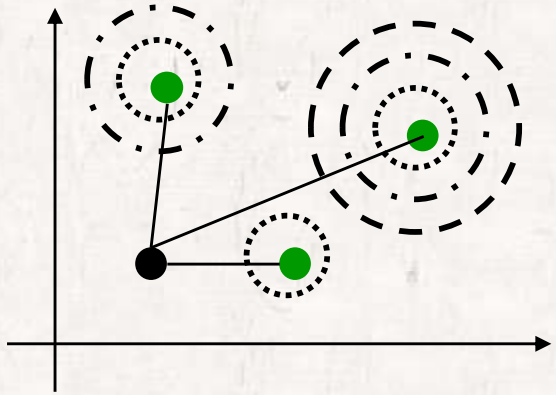


- No biases
- Weights between input and hidden layer can be fixed to 1

Radial-Basis Neural Network

- Structure





- It is a kind of local model based approximation

Learning Algorithm

- Minimize Error Function
  - Error function

$$E = \frac{1}{2}\sum_{i=1}^{n}\left(y_i - N(\mathbf{x}_i)\right)^2$$

  - Centers

$$\Delta m_{ji} = -\eta \frac{\partial E}{\partial m_{ji}}$$

  - Spread

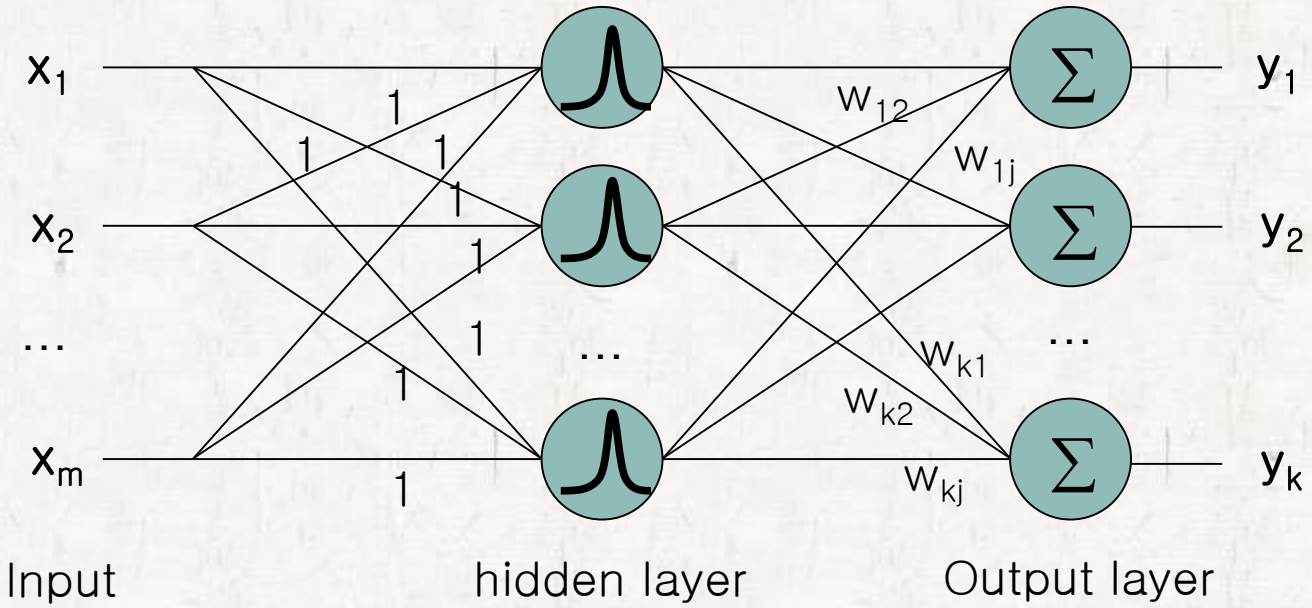
$$\Delta \sigma_j = -\eta \frac{\partial E}{\partial \sigma_j}$$

  - Weights

$$\Delta w_{ji} = -\eta \frac{\partial E}{\partial w_{ji}}$$

Discussion

- For Regression
  - Use a linear output node: Why?



$$y_k = \sum_{j=1}^{H} w_j h_j$$

Are you familiar with this?

Discussion

- For Regression
  - Do I know it ??

$$y_k = \sum_{j=1}^{H} w_j h_j \qquad h_j = \exp\left(-\frac{\left\| \mathbf{x} - \mathbf{m}_j \right\|^2}{2\sigma_j^2} \right)$$

$$y_k = \sum_{j=1}^{H} w_j \exp\left(-\frac{\left\| \mathbf{x} - \mathbf{m}_j \right\|^2}{2\sigma_j^2} \right)$$

  - If x is one dimensional

$$f(x) = w_1 \exp\left(\frac{-(x-m_1)^2}{2\sigma_1^2} \right) + w_2 \exp\left(\frac{-(x-m_2)^2}{2\sigma_2^2} \right) + \cdots + w_H \exp\left(\frac{-(x-m_H)^2}{2\sigma_H^2} \right)$$

Discussion

$$y_k = \sum_{j=1}^{H} w_j \exp\left(-\frac{\|\mathbf{x} - \mathbf{m}_j\|^2}{2\sigma_j^2} \right)$$

- Kernel Regression
  - m and σ are given by experts
  - w is determined by solving linear equations

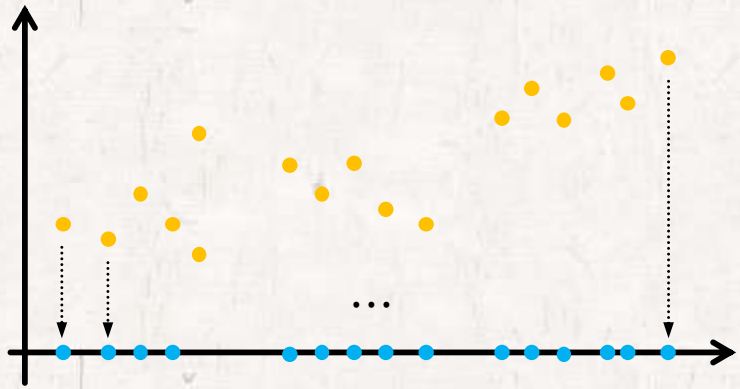
- Radial Basis Neural Network
  - m , σ and w are determined by Gradient Descent Method
  - Another learning algorithm for RBNN with linear output nodes
    - Determine m and σ by unsupervised learning method
    - Determine σ by solving linear equations

Discussion

- Another Learning Algorithm: Example
  - 1 dimensional input and 3 kernels

$$f(x) = w_1 \exp\left(\frac{-(x-m_1)^2}{2\sigma_1^2}\right) + w_2 \exp\left(\frac{-(x-m_2)^2}{2\sigma_2^2}\right) + w_3 \exp\left(\frac{-(x-m_3)^2}{2\sigma_3^2}\right)$$

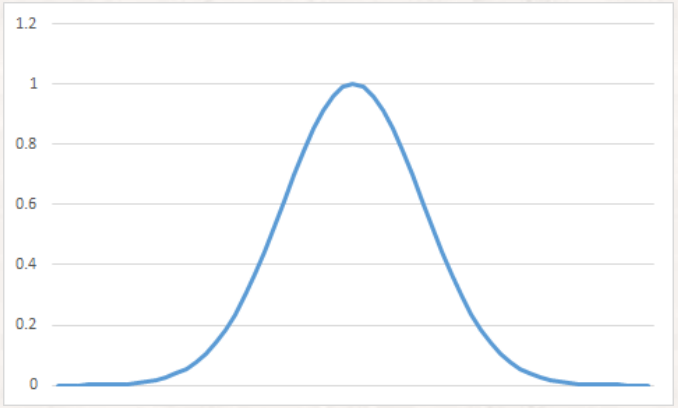
  - Where to place kernels?
    - Drop outputs and cluster inputs into 3 groups
    - The center and variance of each groups are those of each kernel

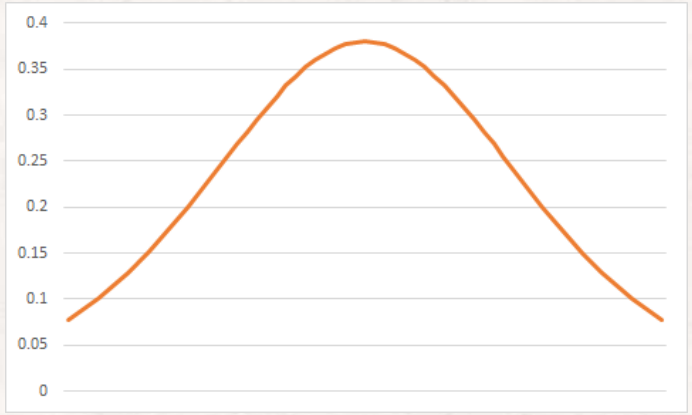


Discussion

- Any Difference between Sigmoid and Radial Basis?
  - Sigmoids can mimic Radial Basis Functions!!

$$\text{Sigmoid}(\mathbf{x}) = \frac{1}{1 - \exp(-\mathbf{x} \cdot \mathbf{w})}$$





$$f(x) = \frac{1}{1-e^{-x}} + \frac{1}{1-e^{x}} - \frac{1}{1-e^{-(x-0.8)}} - \frac{1}{1-e^{x+0.8}}$$